James Letts

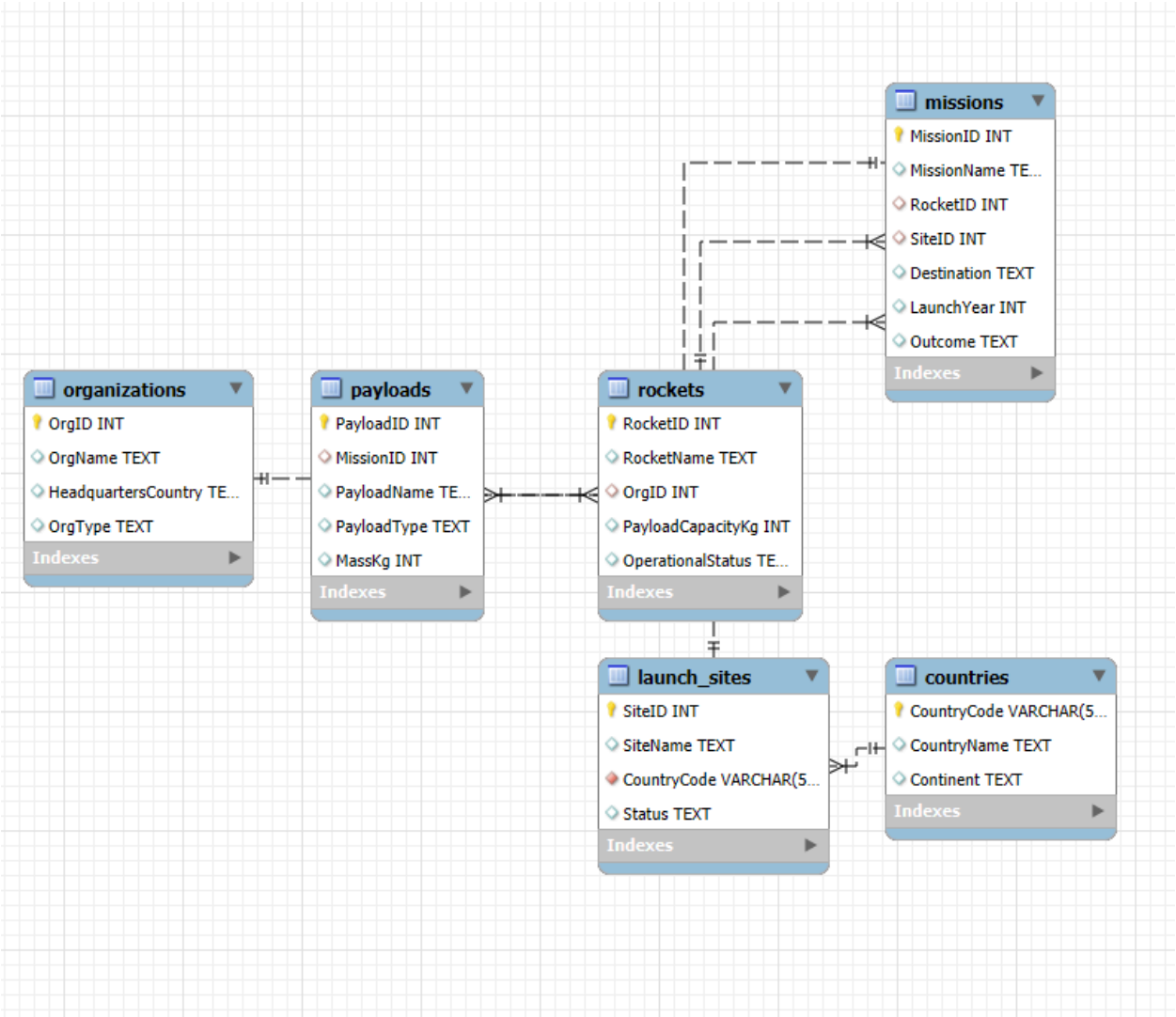
CSC 621 Database Management

Professor Babak Forouraghi

10/26/2026

Assignment 4

- EER Diagram made in MySQL



SQL Query Code:

-- 1.

```
select missions.MissionName, rockets.RocketName, organizations.OrgName, launch_sites.SiteName,
missions.Destination, missions.LaunchYear, missions.Outcome
from missions
join rockets on missions.RocketID = rockets.RocketID
join organizations on rockets.OrgID = organizations.OrgID
join launch_sites on missions.SiteID = launch_sites.SiteID
order by missions.LaunchYear desc, missions.MissionName asc;
```

MissionName	RocketName	OrgName	SiteName	Destination	LaunchYear	Outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

-- 2.

```
select missions.MissionName, rockets.RocketName, payloads.PayloadName, payloads.PayloadType,
payloads.MassKg
from payloads
right join missions on payloads.MissionID = missions.MissionID
left join rockets on missions.RocketID = rockets.RocketID
where missions.LaunchYear = 2024
order by payloads.MassKg desc;
```

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100

-- 3.

```
select launch_sites.SiteName, missions.MissionName, rockets.RocketName, missions.LaunchYear
```

```
from launch_sites
left join missions on launch_sites.SiteID = missions.SiteID
left join rockets on missions.RocketID = rockets.RocketID
group by launch_sites.SiteName,  missions.MissionName, rockets.RocketName, missions.LaunchYear;
```

SiteName	MissionName	RocketName	LaunchYear
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

```
-- 4.
with total_payload_mass as (
select organizations.OrgID as OrgID, sum(payloads.MassKg) as TotalMass
from organizations
join rockets on organizations.OrgID = rockets.OrgID
join missions on rockets.RocketID = missions.RocketID
join payloads on missions.MissionID = payloads.MissionID
group by organizations.OrgID
)
select organizations.OrgName, organizations.HeadquartersCountry, organizations.OrgType,
total_payload_mass.TotalMass
from organizations
join total_payload_mass on organizations.OrgID = total_payload_mass.OrgID
order by total_payload_mass.TotalMass desc;
```

OrgName	HeadquartersCountry	OrgType	TotalMass
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100

```
-- 5.
with total_country_missions as (
select countries.CountryCode as CountryCode, count(missions.MissionID) as CountryMissions
from countries
left join launch_sites on countries.CountryCode = launch_sites.CountryCode
left join missions on launch_sites.SiteID = missions.SiteID
group by CountryCode
),
total_country_mass as (
select countries.CountryCode as CountryCode, sum(payloads.MassKg) as TotalMass
from countries
left join launch_sites on countries.CountryCode = launch_sites.CountryCode
left join missions on launch_sites.SiteID = missions.SiteID
left join payloads on missions.MissionID = payloads.MissionID
group by CountryCode
)
select countries.CountryName, total_country_missions.CountryMissions, ifnull(total_country_mass.TotalMass, 0) as
TotalMass
from countries
join total_country_missions on countries.CountryCode = total_country_missions.CountryCode
join total_country_mass on countries.CountryCode = total_country_mass.CountryCode
order by total_country_mass.TotalMass desc;
```

	CountryName	CountryMissions	TotalMass
▶	United States	3	39492
	Japan	1	6000
	Russia	1	2450
	China	1	1200
	India	1	150
	New Zealand	1	100
	France	0	0
	United Kingdom	0	0